Life-Pulse Guardian

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Why Life-Pulse Guardian? The Problem & Solution

The Problem: Critical Gaps in Disaster Response

- **Time is Critical:** Survival rates for trapped victims plummet by 80% after the first 60 minutes in a disaster.
- **Communication Breakdown:** Traditional cellular networks are often overwhelmed or damaged during emergencies, leading to communication blackouts.
- **Invisible Victims:** Without reliable communication or location, victims in collapsed structures or remote areas become untraceable, delaying crucial rescue operations.
- **Cost & Durability:** Existing tracking devices are typically expensive, bulky, and lack the ruggedness needed to withstand extreme conditions.

The Solution: Life-Pulse Guardian - Reliable, Affordable Safety

- **Network Independent:** Utilizes non-cellular communication for detection, ensuring functionality even when traditional networks fail.
- **Ultra-Affordable:** Designed to be accessible to all, priced at just ₹1,250 (approx. $15 USD).
- **Extended Battery Life:** Provides critical functionality for up to 72 hours, offering a vital window for rescue.
- **Rugged & Resilient:** Engineered with a shockproof design to withstand harsh environments and impacts.
- **Global Reach:** Operates effectively even in signal-deprived zones, making it ideal for remote or disaster-stricken areas.

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Build Components & Costs

1	Arduino Uno (Brain) ₹550 — Main processor
2	ADXL335 Accelerometer ₹250 — Motion detection
3	Conductivity Probes ₹50 — Water sensing
4	16x2 LCD Display ₹250 — Visual interface
5	Panic Button & 100dB Piezo ₹150 — Manual trigger

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Component Functions & Roles

Each component plays a vital role in ensuring the **Life-Pulse Guardian** operates effectively, providing comprehensive safety and alerts.

Arduino Uno

The **brain** of the system, it processes all sensor data and executes control logic for alerts and outputs.

MPU6050 Accelerometer

Detects **immobility** and unusual movement patterns, essential for recognizing falls or prolonged inactivity.

GPS Module

Monitors location accuracy even without netwok

LCD Display

Provides **real-time status updates**, notifications, and alerts directly to the user for immediate awareness.

Panic Button

Allows for a **manual emergency trigger**, enabling users to call for help instantly when needed.

Piezo Buzzer

Emits a **loud 100dB alarm** to attract immediate attention in emergency situations.

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Technical Architecture

Complete system flow from input sensors through Arduino-based processing to output alerts and user feedback.

Inputs

MPU6050 Accelerometer
Panic Button

Arduino Uno

Central processor
Receives and evaluates sensor data

Processing Logic

Filter → Timer → Alarm
Detects critical immobility and triggers escalation

Outputs

LCD Display
Piezo Buzzer
LED Alert

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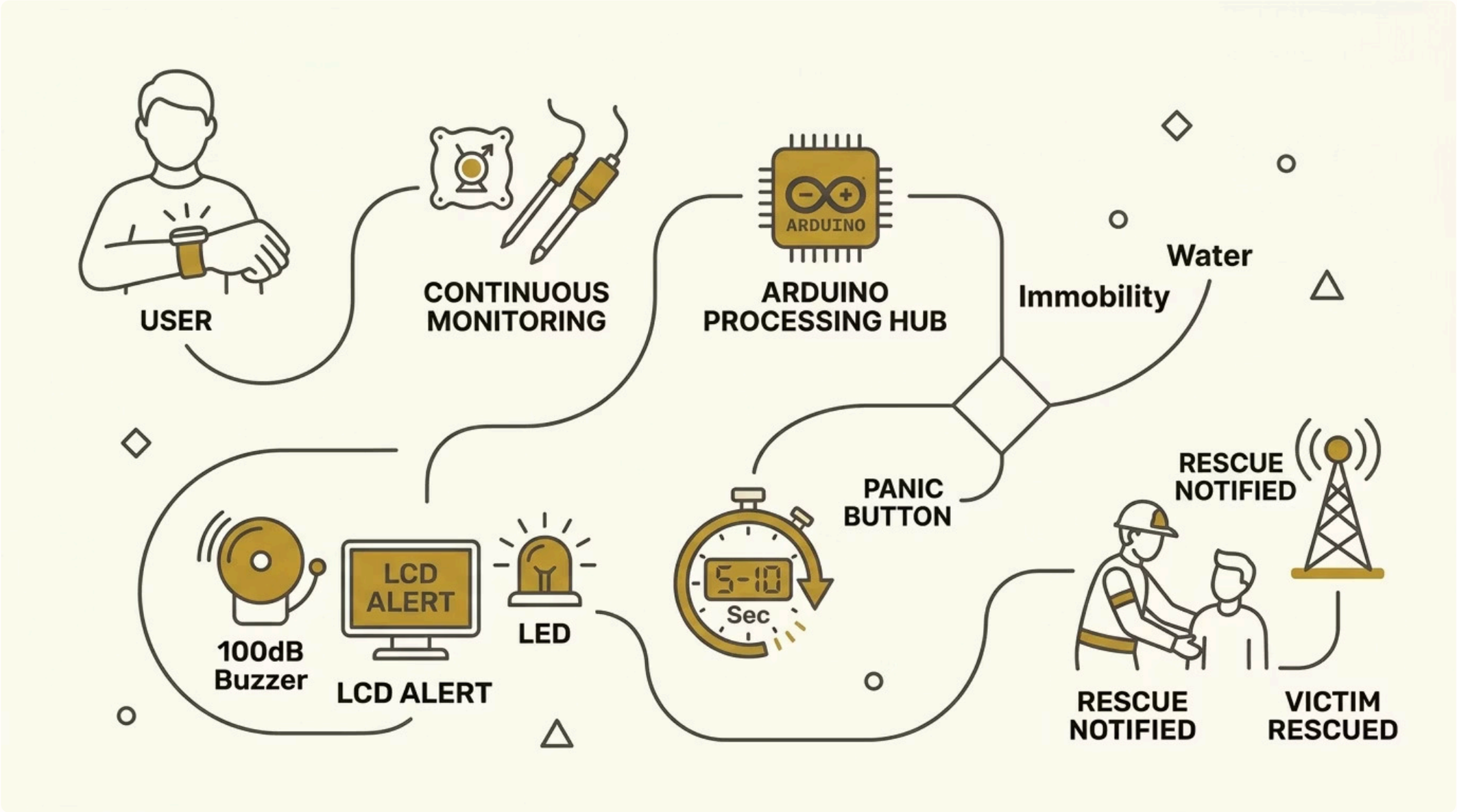
Complete System Process Flow



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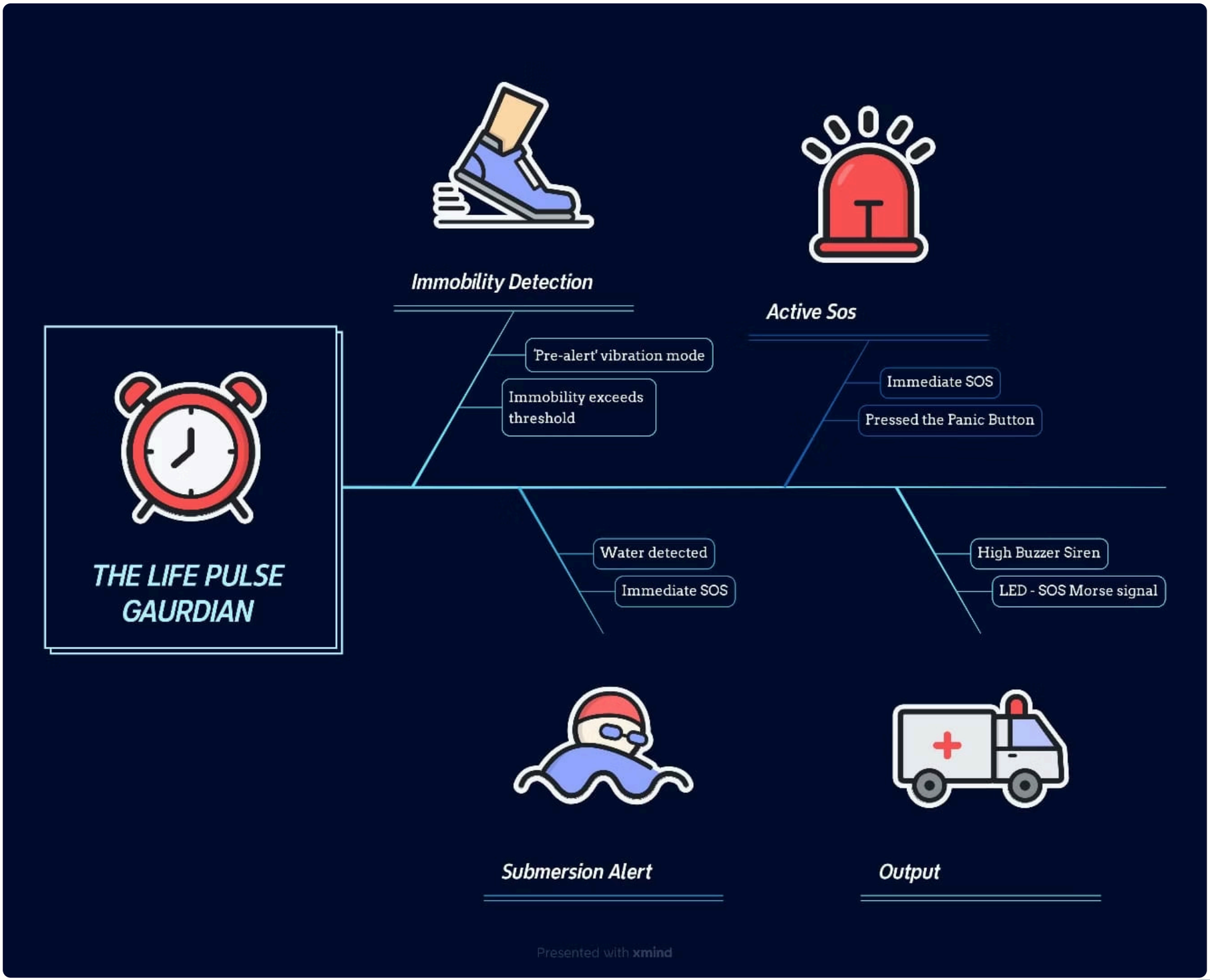

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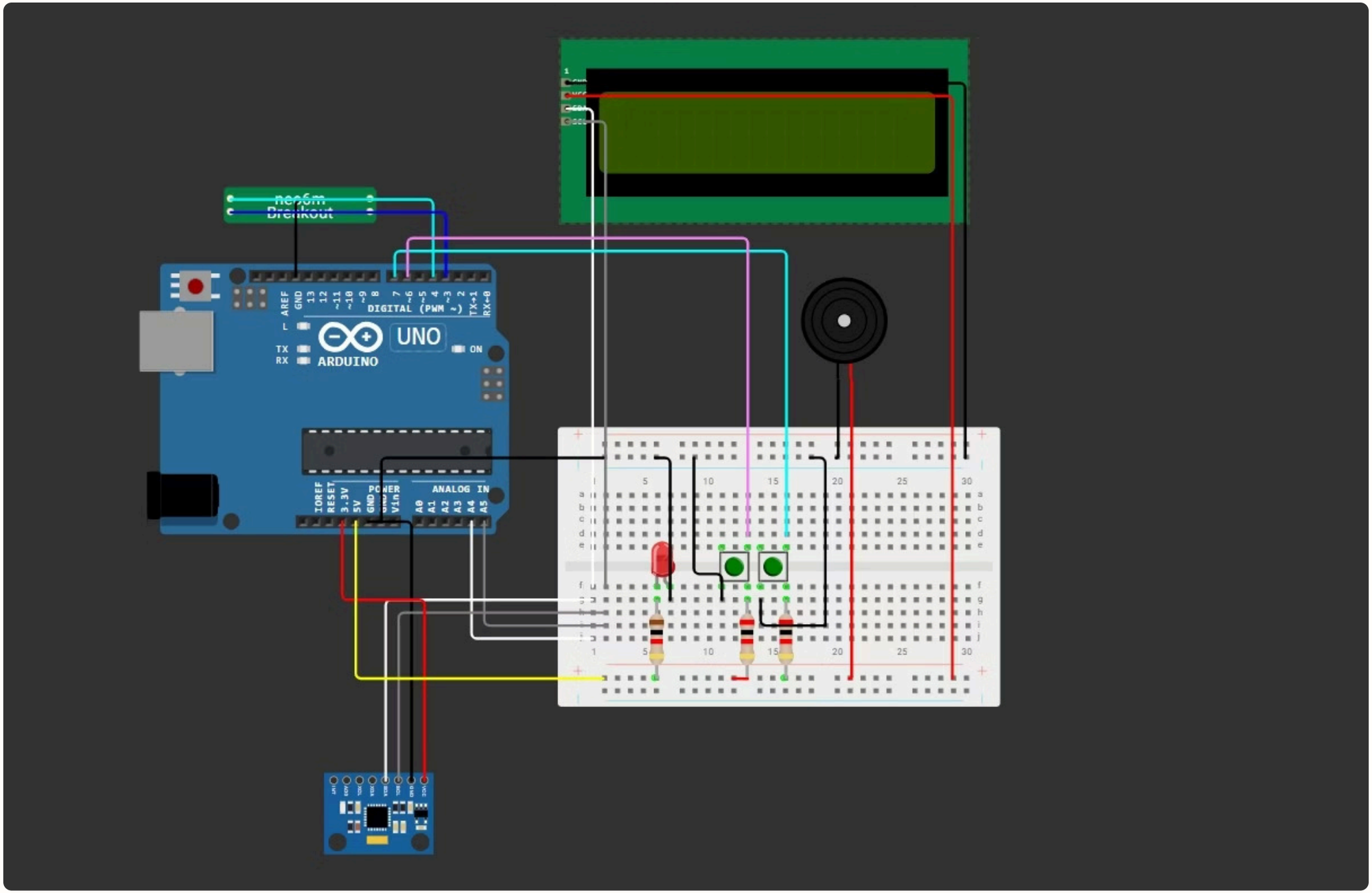

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Key Advantages & Benefits

The Life-Pulse Guardian isn't just a device; it's a game-changer for emergency response, offering unparalleled reliability and accessibility when it matters most.

Saves Lives

Reduces victim location time by **70%**, drastically increasing survival rates in critical scenarios.

Network Independent

Ensures continuous operation without relying on fragile cellular or internet infrastructure during disasters.

Ultra-Affordable

Priced at just **₹1,250** (approx. $15 USD), making vital safety accessible to all rescue teams and individuals.

Extended Battery

Provides up to **72 hours** of continuous operation, a crucial window for search and rescue efforts.

Rugged Design

Engineered to be **shockproof** and disaster-resistant, built to withstand extreme environments and impacts.

Instant Alerts

Features a **100dB alarm** and clear visual indicators to attract immediate attention in emergencies.

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Sustainable Development Goals

Good Health and Well-being

Life-Pulse Guardian contributes by providing timely alerts for individuals in distress, especially those who are immobile or unconscious, improving emergency response and potentially saving lives.

Industry, Innovation and Infrastructure

The project represents innovation in emergency response technology, providing robust infrastructure for personal safety even in areas with unreliable network coverage. It showcases technological advancement in a critical area.

Sustainable Cities and Communities

By enhancing safety and emergency response capabilities, Life-Pulse Guardian helps create safer and more resilient communities, particularly for vulnerable populations or in disaster scenarios.

Climate Action

Life-Pulse Guardian supports climate action by reducing unnecessary rescue operations through early detection, helping minimize carbon footprint through more efficient emergency response, and enabling more sustainable disaster preparedness in climate-vulnerable regions.

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Future Scope

Fire Detection

Using the NTC 100K thermistor, the system can measure temperatures up to 300°C. A safety limit will be set at 100°C, and if the temperature exceeds this threshold, an SOS alert will be automatically triggered to warn users and rescue teams of a potential fire hazard.

Flood Detection & Water Conductivity

Expand conductivity probe technology for comprehensive flood monitoring, water level detection, and early warning systems in flood-prone areas, helping protect communities with faster response and better situational awareness.

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Real-World Application: War & Conflict Zones

In the chaos and destruction of war, every second counts. The Life-Pulse Guardian offers a beacon of hope, providing critical location and safety features where traditional systems fail, profoundly impacting both military and civilian survival rates.

Soldier Safety

Tracks injured or immobilized soldiers on the battlefield, enabling rapid extraction and medical aid and significantly increasing the chances of survival during critical combat situations.

Civilian Protection

Detects civilians trapped in collapsed buildings from airstrikes or explosions, allowing rescue teams to pinpoint locations quickly and effectively, minimizing casualties in urban warfare.

Network Independence

Functions autonomously without relying on traditional cellular networks, which are often overwhelmed, destroyed, or non-existent in war zones, ensuring continuous communication when it matters most.

Cost-Effective Deployment

At an ultra-affordable price of ₹1,250 per unit, the Life-Pulse Guardian can be widely distributed to large numbers of soldiers, rescue teams, and vulnerable populations, making widespread safety accessible.

Life-Saving Potential

Significantly reduces casualty rates by enabling faster location and rescue of wounded personnel or trapped individuals, turning minutes into crucial life-saving opportunities.

Disaster Response

Extends its utility to detect emergent fire hazards and flooding in conflict-affected areas, offering an early warning system against secondary disasters and protecting vulnerable communities.

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